



AI Resume Screening & Recruitment Platform

An **AI-powered Resume Screening & Recruitment Platform** built using the **MERN Stack** and **Generative AI**. The platform helps **job seekers** optimize their resumes for ATS (Applicant Tracking Systems) while enabling **recruiters** to efficiently screen, rank, and manage candidates using AI.



Features



Candidate Module

- Secure JWT Authentication
 - Google OAuth Login (Optional)
 - Upload Resume (PDF/DOCX)
 - Resume Parsing & Information Extraction
 - AI ATS Resume Score (0-100)
 - AI Resume Improvement Suggestions
 - Job Description Matching
 - Missing Skills Detection
 - AI Cover Letter Generator
 - AI Resume Summary Generator
 - AI Interview Question Generator
 - AI Mock Interview
 - Resume Version History
 - Career Recommendation
 - Skill Gap Analysis & Learning Roadmap
 - Resume Analytics Dashboard
 - Download AI Resume Report (PDF)
 - Real-time Notifications
 - Track Applied Jobs
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Recruiter Module

- Recruiter Authentication
- Company Profile Management
- Create & Manage Job Openings
- Upload Job Descriptions
- Bulk Resume Upload
- Automatic Resume Parsing
- AI Candidate Ranking
- ATS-Based Resume Screening
- Resume Filtering
- Candidate Shortlisting

- Interview Scheduling
 - Email Shortlisted Candidates
 - Recruitment Analytics Dashboard
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Admin Module

- Manage Users
 - Manage Recruiters
 - Manage Job Posts
 - View Platform Analytics
 - Manage AI Requests
 - Fraud Detection Monitoring
 - Reports Management
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AI Features

- AI Resume Screening
 - AI ATS Score Generation
 - AI Resume Improvement Suggestions
 - AI Resume Rewrite
 - AI Resume Summary Generator
 - AI Job Description Matching
 - AI Missing Skills Detection
 - AI Skill Gap Analysis
 - AI Learning Roadmap Generator
 - AI Career Recommendation
 - AI Cover Letter Generator
 - AI Interview Question Generator
 - AI Mock Interview Feedback
 - AI Resume Ranking
 - AI Chat Assistant
 - AI Salary Estimation (Future Scope)
 - AI LinkedIn Profile Optimization (Future Scope)
 - AI GitHub Profile Review (Future Scope)
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ATS Analysis

The AI evaluates resumes based on:

- Resume Structure
- Contact Information

- Technical Skills
 - Soft Skills
 - Experience
 - Projects
 - Education
 - Certifications
 - Keywords
 - Grammar
 - Formatting
 - ATS Compatibility
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Recruiter Analytics

- Total Applicants
 - Resume Score Distribution
 - Skills Distribution
 - Experience Distribution
 - Hiring Funnel
 - Candidate Ranking
 - Interview Success Rate
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Security Features

- JWT Authentication
 - Role-Based Access Control (RBAC)
 - Password Hashing (bcrypt)
 - Protected Routes
 - Secure REST APIs
 - Helmet Security
 - CORS Protection
 - Rate Limiting
 - Input Validation
 - File Validation
 - MongoDB Injection Prevention
 - XSS Protection
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Tech Stack

Frontend

- React.js

- Tailwind CSS
- React Router
- Axios
- Redux Toolkit / Context API
- Chart.js / Recharts

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

AI Integration

- OpenAI API / Google Gemini API
- LangChain (Optional)

Authentication

- JWT
- bcrypt
- Google OAuth (Optional)

File Handling

- Multer
- PDF-Parser
- Mammoth (DOCX Parser)
- Tesseract.js (OCR for Scanned Resumes)

Cloud Services

- Cloudinary
- Nodemailer
- Socket.io

Project Structure

```
AI-Resume-Screening/  
|
```

```
├── client/
│   ├── src/
│   ├── public/
│   └── package.json
├── server/
│   ├── config/
│   ├── controllers/
│   ├── middleware/
│   ├── models/
│   ├── routes/
│   ├── services/
│   ├── utils/
│   ├── uploads/
│   └── package.json
├── README.md
└── .env
```

Installation

Clone the Repository

```
git clone https://github.com/your-username/ai-resume-screening.git
cd ai-resume-screening
```

Install Dependencies

Backend

```
cd server
npm install
```

Frontend

```
cd client
npm install
```

Configure Environment Variables

Create a `.env` file inside the `server` folder.

```
PORT=5000

MONGODB_URI=your_mongodb_connection_string

JWT_SECRET=your_jwt_secret

OPENAI_API_KEY=your_openai_api_key

GEMINI_API_KEY=your_gemini_api_key

CLOUDINARY_CLOUD_NAME=your_cloud_name

CLOUDINARY_API_KEY=your_api_key

CLOUDINARY_API_SECRET=your_api_secret

EMAIL_USER=your_email

EMAIL_PASS=your_email_password
```

Run the Project

Start Backend

```
cd server
npm run dev
```

Start Frontend

```
cd client
npm run dev
```

Screenshots

- Home Page

- Candidate Dashboard
 - Recruiter Dashboard
 - Resume Upload
 - ATS Score
 - Resume Analysis
 - Job Matching
 - AI Suggestions
 - AI Mock Interview
 - Analytics Dashboard
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Future Enhancements

- Voice-Based AI Interviews
 - Video Interview Analysis
 - AI Resume Builder
 - Multi-language Resume Analysis
 - Mobile Application
 - Docker & Kubernetes Deployment
 - Microservices Architecture
 - AI Job Recommendation Engine
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Contributing

Contributions are welcome! Feel free to fork the repository, create a feature branch, and submit a pull request.



License

This project is licensed under the MIT License.



Author

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Tech Stack: MERN Stack | AI Integration | REST APIs | MongoDB | React.js | Node.js | Express.js

If you found this project useful, consider giving it a ★ on GitHub.